SCOPE - Activity Recognition Using Temporally Dominant Topic Identification In Forensic Chat Analysis

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Abstract. This paper introduces SCOPE (Semantic Coherence Pruning and Extraction), a framework designed to predict activities or events by identifying temporally dominant topics from large-scale, multi-threaded chat data. Modern investigations face vast volumes of heterogeneous messaging logs across platforms, users, and time, where manual review is infeasible. We posit that detecting a temporally dominant topic over some time can help identify an activity or event of relevance for further analysis. However, there are no prior works that efficiently extract a maximal window length and an associated topic that maintains a high probability of occurrence throughout. This paper formalizes the temporally dominant topic identification problem and presents SCOPE, an efficient algorithm that leverages two-stage pruning to keep search time reasonable for chats over very large windows (e.g., several months or even longer). The overall methodology includes an initial step where the topic probability of each message, and subsequently for each fixed length (unit) window, is computed using a SentenceTransformer model. In the probability pruning phase, windows below the relevance threshold are discarded; in the time pruning phase, contiguous sequences of identified windows are merged and pruned to enforce maximality. Our results show that SCOPE can identify events or activities in chat sequences within a reasonable runtime, and it can serve as a preprocessing module in forensic triage tools.

Keywords: Digital Forensics · Computer Forensics · Pruning · Machine Learning (ML) · Chat Data · Clustering

1 Introduction

Digital forensics constitutes a significant methodological approach for examining digital devices and identifying digital evidence pertinent to cybercrime investigations. The discipline, alternatively designated as computer or cyber forensics, typically encompasses five principal procedural phases: identification of resources and data, extraction of evidence from digital devices, analysis of evidence, documentation of findings, and presentation of evidence in legal proceedings. The requirement for digital forensic investigations appears to be increasing in correlation with the proliferation of criminal activities in digital environments. This increase correlates with the expanded adoption of electronic devices [1]. Consequently, there exists a growing demand for qualified digital forensics investigators. Furthermore, the evolving nature of cybercrime methodologies and continuous technological advancements necessitate adaptive forensic investigation frameworks and procedures to address these incidents effectively [2].

Social media platforms represent a notable source of evidence for forensics investigators. These environments encompass various online platforms, chat rooms, and internet forums [3]. As of April 2024, 62.3% of the global population utilizes social media on a regular basis, with an average daily engagement of 2 hours and 23 minutes [4]. The proliferation and widespread adoption of mobile devices have further contributed to the increased utilization of social media applications [5]. Users engage with diverse social media platforms on mobile devices, including Facebook, Instagram, WhatsApp, Telegram, and YouTube. This engagement results in the inadvertent deposition of substantial personal data on these platforms, thereby establishing mobile devices as critical evidentiary sources in social media-related investigations [6].

Recent research has focused on Online Social Networks (OSNs) for mobile devices. Investigators have conducted numerous studies on various social media applications, examining activity logs, image manipulation techniques, and performing case studies and surveys to monitor criminal activities on Facebook [7][8][9][10]. The popularity of Instagram has similarly prompted extensive research in mobile forensics pertaining to social media applications [11][12][13][14][15]. Additionally, researchers have investigated chat messages and media sharing functionalities on Telegram, Snapchat, and WhatsApp, identifying valuable evidentiary material for forensic examinations [16][17][18][19]. These studies collectively indicate the progression of research in social media applications on mobile devices for forensic purposes.

Despite these advancements, the acquisition of digital evidence from smartphone Instant Messaging applications presents considerable challenges for criminal investigations and judicial proceedings [20]. Users frequently employ multiple instant messaging applications concurrently to communicate on diverse topics. When device(s) are acquired, there are very long periods of time over which the user activity is spread. In many investigations, one is exploring possibilities that will link the owner of the device to the topic of investigation. In such cases, the user's involvement in an activity or presence at a specific location for a certain unknown (and unknown length) time window can be a relevant starting point. This leads to the need for finding, what we term as, *temporally dominant topic*, i.e., a topic and a *maximal time window* where the topic of occurs with a sufficiently high probability throughout this window. Our premise is that identification of such a window and topic can be linked to user's involvement in an activity for that period of time. For example, a person visiting a shopping mall might be discussing the stores and eating joints in the mall in their chats.

Researchers have investigated various machine learning and natural language processing algorithms for text forensics, with topic modeling algorithms gaining particular recognition in digital forensics. The extensive data generated by social media applications on mobile devices is predominantly unstructured, necessitating Natural Language Processing (NLP) techniques that enable the interpretation and extraction of meaning from human language through analysis of grammatical, semantic, and morphological elements [21]. Topic modeling facilitates the organization of textual databases into thematically related groups, enhancing contextual comprehension through the identification of similar textual patterns or structural sequences. However, topic modeling presents challenges including optimization complexities, sensitivity to noise, and instability [22]. Related to, and as a starting point for our work, several topic modeling algorithms have been employed for text mining research, including Probabilistic Latent Semantic Analysis (PLSA) [23][24][25], Latent Dirichlet Allocation (LDA) [26][27][28][29], Correlated Topic Model (CTM) [30], and BERTopic [31] to identify thematic elements within documents. However, no work to date specifically focuses on searching for a time window (of a variable length) where a topic dominates.

This paper formalizes the temporally dominant topic identification problem and present an efficient algorithm, called SCOPE (Semantic Coherence Pruning and Extraction). The essence of our approach involves a two-phase pruning to maintain reasonable search times when dealing with conversations spanning extensive timeframes (such as multiple months or longer). Our methodology begins with calculating topic probabilities for individual messages and subsequently for each fixed-length window segment. The two pruning methods are *Probability Pruning* and *Time Pruning*, which make the task of identifying relevant consecutive subsets of chat messages for any topic more efficiency. We demonstrate the calculation methodology for subset probabilities and establish the efficacy of our algorithm in extracting relevant subsets through pruning methods. This approach enhances computational efficiency compared to traditional brute force algorithms in evidence identification. We illustrate the effectiveness of our methodology through two case studies, highlighting its applicability to social media application investigations on mobile devices.

To demonstrate our approach, we utilized an 8-month-long Chit-Chat dataset containing communications between university students. This dataset encompasses messages from multiple concurrent chatrooms, enabling interpretation of communications across various social media applications. The case studies demonstrate that our proposed approach can identify recent events from the students' chat messages. The primary contributions of this paper are:

– We define the problem of finding temporally dominant topics, specifically pairs of topics and maximal time windows, where a topic occurs above a certain threshold frequency. This is based on the premise that such topics can help identify events or activities, and user's involvement in these, in exploratory forensic analysis.
– We introduce SCOPE, a novel framework that employs semantic embedding and probabilistic pruning to efficiently determine topic and maximal time window pairs without exhaustive search.
– We apply or extend and empirically evaluate three distinct semantic similarity approaches - Cosine Similarity, Hybrid Cosine-KeyBERT and LDA - demonstrating their comparative effectiveness for chat forensics and establishing a methodological foundation for future research in conversational context analysis.
– We formalize a two-phase pruning algorithm that provides substantial computational advantages over the bruteforce approach. Our empirical evaluation demonstrates that the approach significantly reduces the time to process the data while maintaining forensic insight quality.

2 Related Work

Previous research has employed Latent Dirichlet Allocation (LDA) topic modeling for social network analysis in digital forensics. Studies have integrated LDA with feature selection and machine learning to analyze online communication [23] and applied LDA within blockchain frameworks for digital forensics analysis on online social networks [24]. However, these investigations did not focus on user chat messages, limiting their scope to professional communications, statuses, or news on social media. This creates a significant research gap regarding the analysis of rich and informal data found in chat messages.

In the domain of chat data analysis for Digital Forensics, various topic modeling algorithms have been applied. Research has combined LDA topic modeling with itemset mining to address online grooming [26], implemented PLSA for analyzing library chat transcripts [25], and utilized semantic analysis with LDA to identify similar messages across messaging platforms including WhatsApp and Telegram [27]. A common limitation in these studies is their treatment of entire chat logs as single documents, focusing solely on topic identification rather than analyzing individual messages to uncover relationships and understand user motivations.

Beyond Digital Forensics, topic modeling algorithms have been successfully applied for chat data analysis across diverse sectors. In software engineering, LDA has been utilized to identify important discussion topics among developers [28]. In customer service contexts, newer models like BERTopic and Top2Vec have demonstrated superior performance compared to LDA in determining customer intents [31]. Additionally, LDA has been applied to analyze healthcare chat data [29]. These applications demonstrate the versatility of topic modeling algorithms in extracting meaningful insights from chat data across different domains.

The field has also seen integration of various machine learning and NLP techniques with topic modeling for short text data mining. Methodologies combining LDA with Named Entity-Relationship analysis, emotion and sentiment analysis, and linguistic feature extraction have been applied to social media data to understand user mental health [32]. Text mining with machine learning has been used to analyze user experiences with mobile applications [33]. However, these approaches typically focus on specific goals or topics rather than exploring user discussions across multiple concurrent topics.

Research on user profiling through chat message analysis has focused primarily on predicting user intent, satisfaction, personal traits, or responses [34][35][36][37]. These studies predominantly employ machine learning classification methods within specific chat scenarios but lack a comprehensive approach to identify relevant chat messages within temporal boundaries. Given the evolving nature of people's motives and perceptions, connecting all potential chat messages is crucial for accurate user profiling. This necessitates a method that searches for maximal time windows and associated dominant topics.

3 Methodology

Given a corpus C comprising chat messages, and a set T of investigative topics of interest our objective is to identify all *maximally relevant, contiguous message sequences* that exceed a configurable probabilistic relevance threshold P_s. The proposed pipeline applies sentence-level embedding, semantic similarity computation, window-based aggregation, and a two-phase pruning process - *probability pruning* followed by *time pruning* - to progressively refine the solution space and identify the topics and maximal time windows. This formulation enables forensic analysts to efficiently traverse the high-dimensional space of multi-threaded communications, substantially reducing the cognitive load associated with manual inspection. At the same time, it preserves the temporal structure necessary for contextual and evidentiary interpretation. Figure 1 shows an overview of our method.

Notation Let $\mathcal{C} = \{C_1, C_2, \ldots, C_n\}$ denote the set of chatgroups. Each message $m_i \in C_j$ is associated with a timestamp t_i. Let $\{T_1, T_2, \ldots, T_k\}$ be the set of predefined or discovered topics. Let $P_s \in [0, 1]$ denote the minimum topic relevance threshold required to retain a window.

Formal Problem Statement Given a sequence of chat messages and a topic $T_i \in \{T_1, T_2, \ldots, T_k\}$, our objective is to extract all *maximal*, contiguous subsets of time windows that satisfy the probability pruning condition and are not contained within any larger windowed segment. Formally, let $S(T_i)$ denote the set of such segments:

$$S(T_i) = \Big\{ S_j(T_i) \subseteq \{W_1, W_2, \ldots, W_m\} \;\Big|\; P(S_j(T_i)) \geq P_s \text{ and } S_j(T_i) \text{ is maximal} \Big\}$$

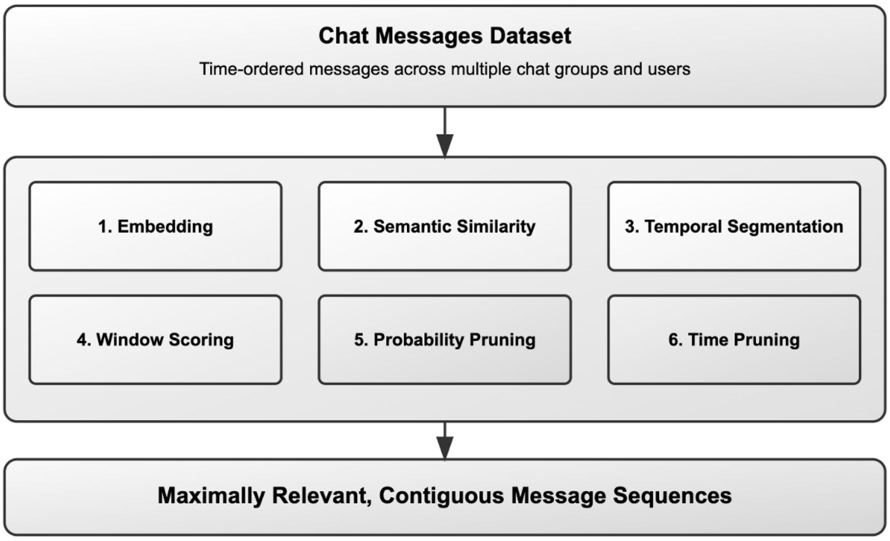

Fig. 1: An Overview of SCOPE Framework

where:

$$S_j(T_i) = \{W_a, W_{a+1}, \ldots, W_b\} \quad \text{with } 1 \leq a \leq b \leq m$$

and $P(S_j(T_i))$ is the average topic probability across all messages in the windows $S_j(T_i)$.

We address the above problem through the following set of steps.

Step 1. Embedding Each message m_i and topic label T_j is embedded into a common semantic space using a SentenceTransformer model (e.g., `all-MiniLM-L12-v2`). The model outputs 384-dimensional dense vectors (where $\mathbf{t}_j$ denotes the embedding vector for topic T_j):

$$\text{Embed}(m_i) \rightarrow \mathbf{w}_i \in \mathbb{R}^{384}, \quad \text{Embed}(T_j) \rightarrow \mathbf{t}_j \in \mathbb{R}^{384}$$

Step 2. Semantic Similarity Estimation To evaluate the effectiveness of different semantic similarity estimation techniques (discussed in Section 3.1) within our temporal and probabilistic pruning pipeline for chat context analysis, we systematically modified Step 2 (Semantic Similarity Estimation) while maintaining all other components of the pipeline constant. This controlled experimental design allows us to isolate the impact of the similarity measure on the overall performance of our forensic chat analysis system. A conceptual contribution of our work is the principled transformation of semantic similarity scores into proper probability distributions (Probabilistic Framework Using Softmax Transformation). While similarity measures quantify the semantic relatedness between messages and topics, they do not inherently represent probabilities. Therefore, we employ the softmax function to transform similarity scores into a valid probability distribution over topics:

$$P(T_j \mid m_i) = \frac{\exp(\beta \cdot \text{sim}(m_i, T_j))}{\sum_{k=1}^{K} \exp(\beta \cdot \text{sim}(m_i, T_k))} \tag{1}$$

where β is a temperature parameter controlling the sharpness of the distribution (with higher values creating sharper distributions). This transformation ensures that: (1) all probabilities are in the range $[0, 1]$, (2) probabilities sum to 1 across all topics for a given message, and (3) the relative ordering of similarity scores is preserved.

Step 3. Temporal Segmentation Each chatgroup $C_i \in \mathcal{C}$ is segmented into a sequence of non-overlapping time windows:

$$\{W_1, W_2, \ldots, W_m\}, \quad \text{where } W_k \subseteq C_i$$

Each window W_k contains a subset of messages that fall within a fixed temporal interval (e.g., 5 minutes).

Step 4. Window Scoring The topic relevance of a window W_k for a given topic T_j is computed by averaging the message-level probabilities:

$$P(T_j \mid W_k) = \frac{1}{|W_k|} \sum_{m_i \in W_k} P(T_j \mid m_i)$$

Step 5. Probability Pruning In the first pruning phase, we discard any window W_k that does not meet the minimum topic relevance threshold:

$$W_k \in PP(T_j) \iff P(T_j \mid W_k) \geq P_s$$

This step significantly reduces the search space by eliminating windows that are unlikely to be relevant, ensuring that only semantically meaningful segments progress to the next stage.

Step 6. Time Pruning Next, we enforce a temporal maximality condition to avoid redundant overlapping segments. A sequence of windows $\{W_a, W_{a+1}, \dots, W_b\} \subseteq PP(T_j)$ is maximal if:

- The average topic probability across the sequence meets or exceeds the threshold:

$$P(T_j \mid \{W_a, \dots, W_b\}) \geq P_s$$

- Extending the sequence with adjacent windows would reduce the average below P_s:

$$P(T_j \mid \{W_{a-1}, \dots, W_b\}) < P_s \quad \text{or} \quad P(T_j \mid \{W_a, \dots, W_{b+1}\}) < P_s$$

3.1 Semantic Similarity Approaches

We implemented and compared three distinct approaches for measuring semantic relatedness between chat messages and investigative topics:

Approach #1. Cosine Similarity (Baseline) Our baseline approach computes the cosine similarity between message embeddings $\mathbf{w}_i$ and topic embeddings $\mathbf{t}_j$ generated using a SentenceTransformer model:

$$\text{sim}(m_i, T_j) = \frac{\mathbf{w}_i \cdot \mathbf{t}_j}{\|\mathbf{w}_i\| \cdot \|\mathbf{t}_j\|} \tag{2}$$

This similarity score is then transformed into a probability using the softmax function as described in our probabilistic framework. The approach captures the semantic orientation of messages in the embedding space but may lack sensitivity to key topical terms.

Approach #2. Hybrid Cosine-KeyBERT Approach The second approach integrates KeyBERT [38] to identify and leverage topic-specific keywords, combining it with cosine similarity before applying the softmax transformation:

$$\text{sim}(m_i, T_j) = \frac{\mathbf{w}_i \cdot \mathbf{t}_j}{\|\mathbf{w}_i\| \cdot \|\mathbf{t}_j\|} \cdot \text{KeywordRelevance}(\text{KeyBERT}(m_i), T_j) \tag{3}$$

KeyBERT(m_i) extracts the most relevant keywords from message m_i, and KeywordRelevance computes how relevant these extracted keywords are to topic T_j.

Approach #3. Latent Dirichlet Allocation (LDA) The third approach employs LDA [39] to model the distribution of latent topics across messages:

$$\text{sim}(m_i, T_j) = \theta_{i,j} \tag{4}$$

where θ_i represents the topic distribution for message m_i, and $\theta_{i,j}$ is the probability of topic T_j occurring in message m_i. LDA naturally produces a probability distribution (as $\sum_{j=1}^{K} \theta_{i,j} = 1$), yet we still apply the softmax transformation to ensure consistency with other approaches and to allow temperature-based calibration. For LDA implementation, we aggregated messages into larger temporal segments to address the sparsity challenge presented by short chat messages. We optimized the number of topics k using perplexity and coherence metrics.

Algorithm 1 Algorithm used in SCOPE Framework

Require: Set of chatgroups $\mathcal{C} = \{C_1, C_2, \ldots, C_n\}$, topics $\{T_1, T_2, \ldots, T_k\}$, threshold P_s, window duration Δt
Ensure: Set of maximal topic-relevant segments $S(T_j)$ for each topic T_j
1: **for** each message $m_i \in C_j$ where $C_j \in \mathcal{C}$ **do**
2: $\mathbf{w}_i \leftarrow \text{Embed}(m_i)$
3: **end for**
4: **for** each topic T_j **do**
5: $\mathbf{t}_j \leftarrow \text{Embed}(T_j)$
6: **end for**
7: **for** each message m_i and topic T_j **do**
8: Calculate $\text{sim}(m_i, T_j)$ based on chosen approach
9: $P(T_j|m_i) \leftarrow \frac{\exp(\beta \cdot \text{sim}(m_i,T_j))}{\sum_{l=1}^{k} \exp(\beta \cdot \text{sim}(m_i,T_l))}$
10: **end for**
11: **for** each chatgroup $C_j \in \mathcal{C}$ **do**
12: Segment C_j into windows $\{W_1, W_2, \ldots, W_m\}$ of duration Δt
13: **end for**
14: **for** each topic T_j **do**
15: // **Window Scoring**
16: **for** each window W_l **do**
17: $P(T_j|W_l) \leftarrow \frac{1}{|W_l|} \sum_{m_i \in W_l} P(T_j|m_i)$
18: **end for**
19: // **Probability Pruning**
20: $PP(T_j) \leftarrow \{W_l : P(T_j|W_l) \geq P_s\}$
21: // **Time Pruning**
22: $S(T_j) \leftarrow \emptyset$, curr $\leftarrow \emptyset$
23: **for** $l = 1$ to m **do**
24: **if** $W_l \in PP(T_j)$ **then**
25: **if** curr $= \emptyset$ **then**
26: curr $\leftarrow \{W_l\}$
27: **else**
28: temp $\leftarrow$ curr $\cup \{W_l\}$
29: $P(T_j|\text{temp}) \leftarrow \frac{1}{|\text{temp}|} \sum_{W_r \in \text{temp}} P(T_j|W_r)$
30: **if** $P(T_j|\text{temp}) \geq P_s$ **then**
31: curr $\leftarrow$ temp
32: **else**
33: $S(T_j) \leftarrow S(T_j) \cup \{\text{curr}\}$, curr $\leftarrow \{W_l\}$
34: **end if**
35: **end if**
36: **else if** curr $\neq \emptyset$ **then**
37: $S(T_j) \leftarrow S(T_j) \cup \{\text{curr}\}$, curr $\leftarrow \emptyset$
38: **end if**
39: **end for**
40: **if** curr $\neq \emptyset$ **then** $S(T_j) \leftarrow S(T_j) \cup \{\text{curr}\}$
41: **end if**
42: **end for**
43: **return** $\{S(T_1), S(T_2), \ldots, S(T_k)\}$

3.2 Algorithm Implementation

Algorithm 1 presents the complete algorithm used in SCOPE framework implementation. The algorithm takes as input a corpus of chat messages organized into chatgroups $\mathcal{C}$, a set of investigative topics, a minimum probability threshold P_s, and a fixed window duration Δt. It proceeds systematically through the six steps of our pipeline: embedding messages and topics into a common semantic space (lines 1-6), computing similarity scores and transforming them into probabilities (lines 7-9), performing temporal segmentation (lines 10-12), scoring each window based on topic relevance (lines 14-16), applying probability pruning to filter low-relevance windows (lines 18-22), and executing time pruning to identify maximal contiguous segments (lines 24-41). The algorithm outputs a set of maximal time windows and associated topics that satisfy our formal problem statement, efficiently reducing the search space through our two-phase pruning approach.

3.3 Implementation & Parameter Choices

As previously mentioned, all embeddings were generated using the `all-MiniLM-L12-v2` model from the SentenceTransformers library, producing 384-dimensional vectors. Topic Selection Rationale: We identify all possible consecutive subsets of chat snippets for all users across 21 topics, namely: *'News', 'Research', 'Technology', 'Travel', 'Personal', 'Education', 'Career', 'Health', 'Sports', 'Vacation', 'Movie', 'Entertainment', 'Book', 'Event', 'Food', 'Politics', 'Finance', 'Relationships', 'Religion', 'Immigration', 'Fantasy'*. The selection of these topics is essential for this research as they provide the semantic foundation upon which the entire SCOPE framework operates. Without these predefined topic labels, the framework would face critical technical limitations: there would be no semantic anchor points for computing similarity scores in the embedding space, no basis for forming the probability distributions crucial to the softmax transformation, and no mechanism for the probability and time pruning phases that depend on topic relevance calculations. From an investigative perspective, digital forensics analysts would lack the ability to target their analysis toward specific areas of concern, losing the forensic focus needed to identify maximally relevant, contiguous message sequences that exceed probability thresholds. LDA was implemented using the gensim library with topic numbers ranging from 1-21.

The probability threshold P_s was set to 0.07 for all experiments, based on preliminary testing that showed this value provided an optimal trade-off between precision and recall. (Section 4.2 shows our experimental results by varying the threshold.) Our implementation uses hourly intervals for temporal segmentation, providing a broader contextual view while maintaining computational efficiency. This granularity proved effective for capturing topic evolution in extended conversations while reducing computational complexity. After empirical testing, we set $\beta = 1.0$ to create sufficiently discriminative probability distributions.

In addition to topic and temporal segmentation, we incorporated user-specific analysis, processing chat segments on a per-user basis. This additional dimension allows the system to detect topic-relevant contributions from specific participants and identify user-level patterns of interest to forensic investigators. The user-specific segmentation is particularly valuable in multi-party conversations where different participants may exhibit varying degrees of involvement with topics of interest.

3.4 Computational Advantages Over Brute Force Approach

Our proposed heuristic pipeline offers significant computational advantages over a brute force baseline approach. While the baseline method executes only Steps 1-4 (embedding, similarity estimation, temporal segmentation, and window scoring), it must evaluate all possible consecutive subsets of temporal windows - a computational complexity of $O(m^2)$ for m windows. For large-scale investigations with thousands of messages across multiple chat groups, this quadratic growth becomes prohibitively expensive, often requiring hours or days of processing time. In contrast, our two-phase pruning strategy (Steps 5-6) progressively reduces the search space by first applying probability pruning to eliminate windows with insufficient topic relevance ($P(T_j \mid W_k) < P_s$), and then enforcing temporal maximality conditions to identify only the most informative, non-redundant segments. Empirical testing reveals that this approach typically reduces the computational load while maintaining equivalent or superior forensic insight, making it particularly valuable for time-sensitive digital investigations where investigator cognitive load must be minimized.

3.5 Relation to Frequent Itemset Mining

Our methodology for temporal and probabilistic pruning of evolving chat contexts extends principles from frequent itemset mining to address the unique challenges of digital forensics investigations. While traditional frequent itemset mining algorithms like Apriori [40] and FP-Growth [41] operate on discrete items with binary presence/absence indicators across unordered transaction sets, our approach innovates by working with continuous probability scores across temporally ordered message sequences. This fundamental shift enables forensic analysts to identify maximally relevant, contiguous conversation segments that exceed configurable probabilistic relevance thresholds - a critical capability when investigating multi-threaded communications. Our two-phase pruning strategy, combining probability and temporal pruning, shares conceptual similarities with the candidate generation and testing phases in traditional frequent pattern mining [42], but adapts these principles to preserve the temporal context essential for evidentiary interpretation in digital forensics.

4 Experiments

4.1 Experimental Settings

Hardware and Software Configurations All experiments were conducted on a MacBook Pro M1 Max with the following specifications: Total Number of Cores: 10 (8 performance cores and 2 efficiency cores), Memory: 64 GB, System Firmware Version: 10151.121.1, and OS Loader Version: 10151.121.1. The software environment used includes Anaconda version 2024.06-1.

Dataset This study utilizes an 8-month dataset of university student chat conversations from the ChitChat corpus [43]. The resulting collection contains 258,146 lines of courteous English-language exchanges with timestamps, adding up to 2,478,521 words with no emojis present in the messages. The dataset encompasses 1,315 users across 7,168 chatrooms, with individual users often participating in multiple rooms simultaneously. All messages were collected between April 10 and November 16, 2018. We applied standard text normalization techniques including tokenization, whitespace trimming, and case normalization. We further processed the text by converting all words to lowercase, removing stop-words, and implementing WordNetLemmatizer and SpellChecker for word correction.

While the ChitChat dataset does not originate from multiple messaging platforms, it reflects key characteristics of multi-app environments, such as concurrent participation in multiple chatrooms, asynchronous message flows, and fragmented topical discussions. Thus, it serves as a valid surrogate for evaluating semantic-driven forensic methods intended for cross-platform chat evidence analysis.

4.2 Evaluation Results

Performance of Semantic Similarity Approaches in SCOPE Our methodology has considered three semantic similarity approaches (Cosine Similarity, Hybrid Cosine-KeyBERT and LDA). Given the time-sensitive nature of digital forensic investigations and the practical constraints of processing large-scale chat datasets, we evaluated computational efficiency alongside semantic accuracy. Consequently, we elected to conduct a comprehensive experimental evaluation focusing specifically on Cosine Similarity and LDA approaches, which provided an optimal balance between semantic fidelity (how accurately a method preserves and represents the true meaning and semantic relationships in the text data being analyzed) and processing efficiency.

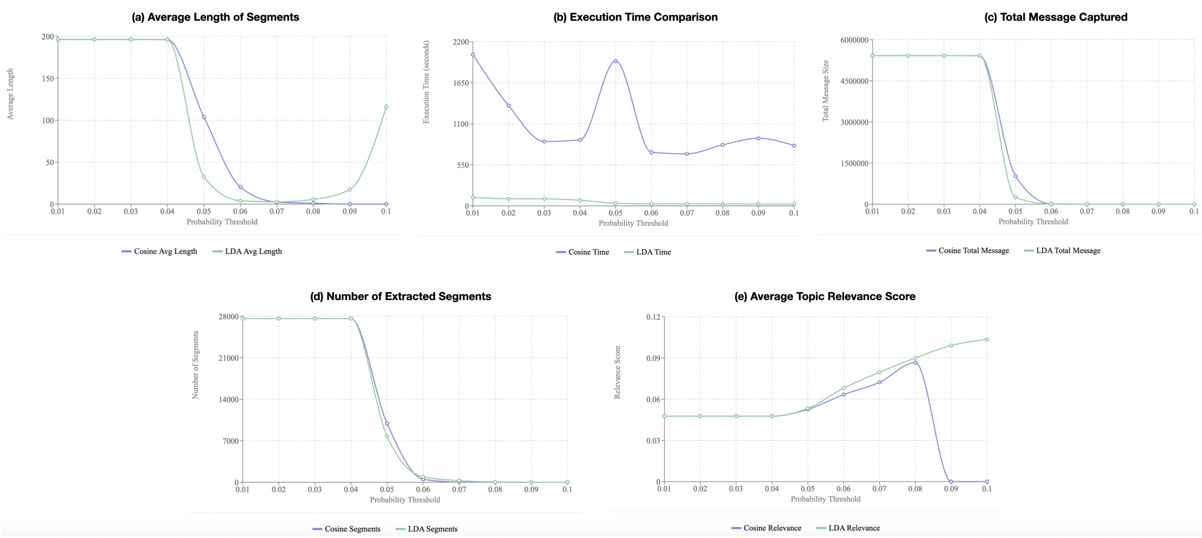

Fig. 2: SCOPE Comparative Evaluations at Various Probability Thresholds

Our evaluation, shown in Figure 2, revealed substantial performance differences between Cosine similarity and LDA topic modeling approaches used to compute the semantic similarity on the entire dataset. While both methods extracted similar numbers of segments at lower thresholds (approximately 27,615 segments at 0.01 - 0.04 thresholds), LDA demonstrated remarkable efficiency advantages, processing data approximately 20 times faster than Cosine similarity (53.39 vs. 1100.39 seconds on average). Furthermore, LDA maintained effectiveness at higher probability thresholds (0.08 - 0.10), and consistently achieved superior topic relevance scores (0.07 vs. 0.05 on average). As probability thresholds increased above 0.05, LDA continued extracting meaningful segments with increasing relevance scores, while Cosine extraction rapidly deteriorated, suggesting LDA offers greater robustness across varying threshold parameters. These findings strongly indicate that LDA represents a better approach for our heuristic pipeline and large-scale topic modeling applications where both processing efficiency and extraction quality are critical considerations.

Comparison with Bruteforce Technique Our SCOPE framework's experimental analysis compared three approaches: (1) Heuristic Pipeline using LDA, (2) Heuristic Pipeline using Cosine similarity, and (3) Bruteforce technique using LDA (without any pruning mechanism).

Short-term Analysis (1-10 days): Table 1 summarizes the execution time statistics across all probability thresholds (0.01 - 0.1) and various time windows (1-10 days) tested. We measured execution time as the system time between the initiation and completion of each algorithm, implemented using Python's time module functions.

Table 1: Short-term Analysis - Execution Time Performance (seconds)

Method	**Mean**	**Min**	**Max**	**Median**
SCOPE (LDA)	7.16	0.07	46.00	1.72
SCOPE (Cosine)	74.14	2.57	421.60	39.06
Bruteforce	402.19	3.42	1550.03	183.08

The performance ratios between methods reveal dramatic efficiency differences:

- Bruteforce/LDA Ratio: 56.17x
- Bruteforce/Cosine Ratio: 5.43x
- Cosine/LDA Ratio: 10.35x

These results conclusively demonstrate LDA's efficiency, being over 10 times faster than Cosine and more than 56 times faster than the Bruteforce approach without pruning.

Comprehensive Analysis (Including Extended Time Periods): Table 2 summarizes the execution time statistics across all probability thresholds (0.01 - 0.1) and various time windows (1-10 days, 1-month, 2-months, 3-months and 7.2-months) tested.

Table 2: Comprehensive Analysis - Execution Time Performance (seconds)

Method	**Mean**	**Min**	**Max**	**Median**
SCOPE (LDA)	17.75	0.07	112.63	8.31
SCOPE (Cosine)	203.34	2.57	2027.39	105.12
Bruteforce	6162.62	3.42	34018.98	507.96

The performance ratios between methods reveal dramatic efficiency differences:

- Bruteforce/LDA Ratio: 347.23x
- Bruteforce/Cosine Ratio: 30.31x
- Cosine/LDA Ratio: 11.46x

Our visualizations of execution time across probability thresholds (0.01 - 0.10) and time windows (1–10 days, 1-month, 2-months, 3-months and 7.2-months) shown in Figure 3 and Figure 4. further confirmed that LDA maintains its performance advantage across all tested parameters, making it the optimal choice for our SCOPE semantic similarity pipeline.

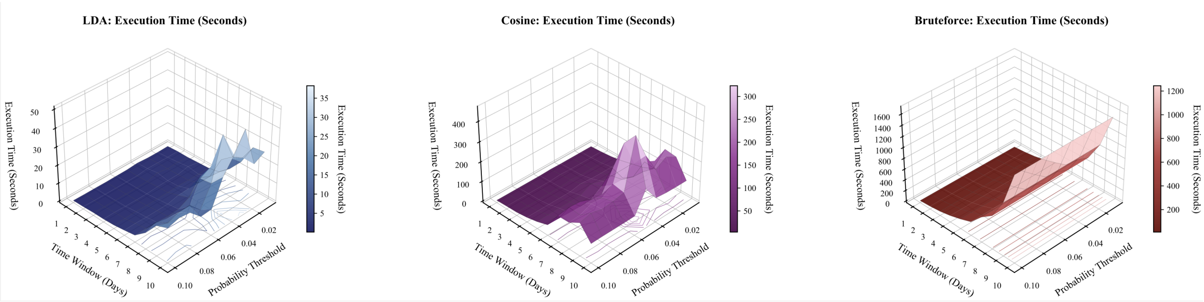

Fig. 3: Short-term Analyis - Execution Time - Performance Comparison

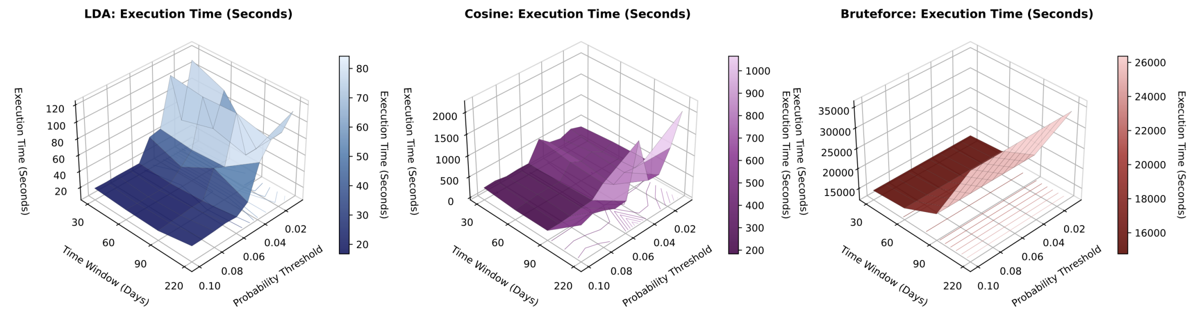

Fig. 4: Comprehensive Analysis - Execution Time - Performance Comparison

4.3 Comparison with BERTopic

To further evaluate SCOPE's effectiveness, we compared our approach with BERTopic [44], a popular topic modeling technique that leverages BERT embeddings and clustering algorithms. We applied BERTopic to the same ChitChat dataset, which generated a total of 6,109 topics: {Topic 0: hello, Topic 1: hi, Topic 2: you, Topic 3: haha, Topic 4: hey, Topic 5: students, Topic 6: computer, Topic 7: courses, Topic 8: medicine, Topic 9: AI, ...}.

The BERTopic results highlight several limitations when applied to forensic chat analysis. First, the model generated an excessive number of topics (6,109), making it challenging for investigators to focus on relevant content. Second, many of the highest-ranked topics (e.g., Topics 0-4) consist primarily of conversational elements like greetings and expressions rather than substantive content relevant to investigative purposes. Third, and most critically, BERTopic lacks the temporal dimension that SCOPE specifically addresses. While BERTopic successfully clusters semantically similar terms, the absence of temporal context prevents investigators from identifying specific time windows where particular activities or events might have occurred. For instance, Topics 6 (computer), 8 (medicine), and 9 (artificial intelligence) potentially relate to educational or professional activities, but BERTopic provides no mechanism to associate these topics with specific temporal segments. In contrast, SCOPE's two-phase pruning methodology explicitly identifies maximally relevant, contiguous message sequences that exceed configurable probabilistic relevance thresholds for specific investigative topics. This approach enables forensic analysts to isolate temporal windows containing evidence-rich exchanges without the cognitive overhead of manually reviewing thousands of algorithmically generated topics.

5 Real-World Investigative Value of SCOPE

In our analysis using the ChitChat dataset described previously, we investigated multiple date ranges using a human-in-the-loop verification approach to validate our findings. Our first case examined activity on April 18, 2018, where User-68 engaged in Chatroom6780 from 22:04:49 to 23:59:42 discussing the 'Movie' topic, notably mentioning "Black Panther" as a film they had heard positive reviews about; human verification confirmed this was a recent reference, as the film was released on February 16, 2018 in the United States [45]. Our second case focused on the period between September 26-October 5, 2018, where human analysts verified that User-253 maintained an unusually prolonged three-day discussion about 'Sports,' specifically referencing the suspension of 13 football players - a news item that

Table 3: Comparative Analysis of Forensic Text Analysis Methods

Feature	**Fundamental Keyword Search Technique**	**LDA Topic Modeling**	**Entity Recognition**	SCOPE
Processing Approach	Rule-based matching of predefined terms	Probabilistic distribution of topics across corpus	Identification of named entities (people, places, organizations)	**SentenceTransformer embeddings with sequential probability and time-based pruning**
Handling of Large Datasets	Linear scaling with message volume; becomes unmanageable with datasets >10,000 messages	Requires extensive preprocessing; degraded performance with short, informal messages	Effective for structured text but struggles with informal communications	**Tested on a dataset containing 258,146 messages across 7,168 chatrooms**
Temporal Analysis	No native temporal analysis capabilities	Limited time-series analysis	No intrinsic time-based analysis	**Segments conversations into fixed-length temporal windows**

was independently confirmed to have been reported on September 1, 2018 [46], demonstrating how our methodology effectively identifies temporally relevant discussions that correlate with real-world events.

Overall, SCOPE's embedding-based approach offers distinct advantages for processing heterogeneous chat data through its two-phase pruning architecture, particularly in temporal segmentation and handling of large message volumes. This comparison focuses on features most relevant to forensic chat analysis rather than attempting to supplant general-purpose forensic platforms. Real-world cases such as cryptocurrency fraud investigations[47,48] and child exploitation cases [49,50], as well as many others, and frequently involve analyzing large volumes of cross-platform communications, which can be supported through our work. Table 3 presents a feature-by-feature comparison between SCOPE and three widely-used text analysis approaches in digital forensics.

6 Formal Guarantees and Complexity Analysis

We briefly state the correctness and complexity properties of our pruning pipeline.

Proposition 1 (Correctness) Given a sequence of time-ordered chat windows and a relevance threshold P_s, Algorithm 1 correctly identifies all *maximal*, topic-relevant, temporally contiguous sequences for each topic $T_j \in \{T_1, \ldots, T_k\}$.

Proof Sketch The algorithm first filters out all windows below the threshold using probability pruning. Then, for the remaining windows, it iteratively aggregates sequences until appending a new window would drop the average below P_s. This ensures both threshold satisfaction and temporal maximality as per the formal problem definition.

Proposition 2 (Complexity) Let there be n time windows, k topics, and assume the similarity computation for a single window-topic pair requires K time units. The worst-case time complexity (when almost no windows are pruned) of the algorithm is:

$$O(n^2 \cdot k \cdot K)$$

7 Future Work - Proposed Integration into Forensic Platforms

We propose integrating SCOPE into open-source forensic platforms like Autopsy as a topic-aware triage module for messaging artifacts. SCOPE's core components (Embedding Engine, Windowing & Scoring, Probability Pruner, and Time Pruner) would leverage existing APIs to process normalized chat data,

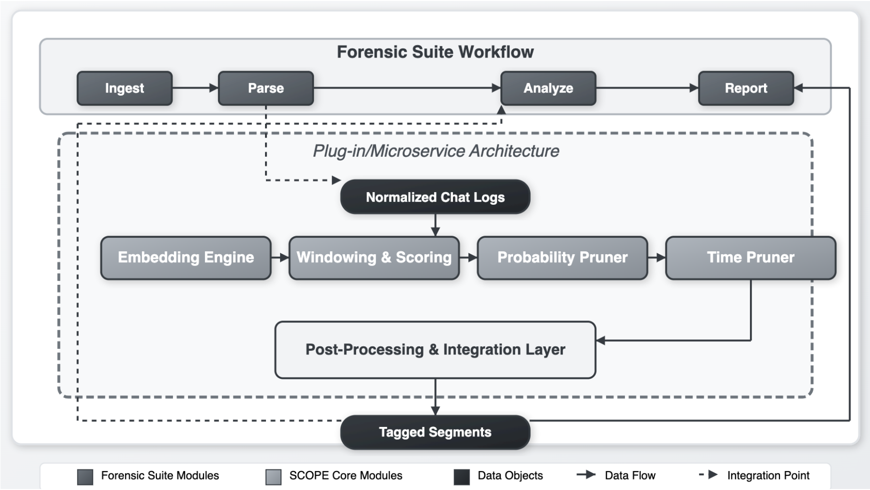

Fig. 5: A Futuristic Architectural Framework for SCOPE Integration in Forensic Platforms

calculating topic probabilities for configurable time windows. After standard ingestion and parsing of chat logs into normalized storage (e.g., Autopsy SQLite database or Cellebrite JSON export), SCOPE would process raw messages, compute embeddings, segment by fixed-duration windows, and calculate topic probabilities. Investigators would access SCOPE through a dedicated interface allowing parameter adjustment (relevance threshold P_s, window duration Δt, embedding model selection). Results would appear as interactive timeline segments with persistent annotations maintaining chain-of-custody compliance. Development plans include a fully integrated plugin or service for Autopsy, with future research focusing on performance benchmarking, usability testing, and support for dynamic windows and specialized embedding models. A visual of the proposed architecture is shown in Figure 5. Moreover, future research could extend our methodology to enhance digital forensics through integration with anomaly detection and behavioral profiling techniques.

8 Conclusion

Our SCOPE framework introduces an efficient approach to identifying temporally dominant topics in forensic chat analysis. The cornerstone of our method is a two-stage pruning strategy that makes analysis feasible for extensive chat windows spanning months or longer. We begin by computing topic probabilities for individual messages using SentenceTransformer embeddings, then aggregate these into fixed-length windows before applying probability and time pruning sequentially. The comparative evaluation demonstrates LDA's substantial advantages over Cosine Similarity, processing data several times faster while maintaining superior topic relevance scores across varying thresholds. Validation on an 8-month dataset confirmed SCOPE's ability to identify meaningful conversational segments while preserving crucial temporal context, significantly reducing investigator cognitive load. Future work will extend this framework to develop comprehensive user behavioral profiles for anomaly detection and seamlessly integrate with existing forensic platforms for field deployment.

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